

tional encodings $\mathbf{R} \in \mathbb{R}^{L_{\max} \times d}$, where the i -th row $\mathbf{R}_i$ indicates a relative distance of i between two positions. By injecting the relative distance dynamically into the attention score, the query vector can easily distinguish the representations of $x_{\tau,j}$ and $x_{\tau+1,j}$ from their different distances, making the state reuse mechanism feasible. Meanwhile, we won't lose any temporal information, as the absolute position can be recovered recursively from relative distances.

Previously, the idea of relative positional encodings has been explored in the context of machine translation (Shaw et al., 2018) and music generation (Huang et al., 2018). Here, we offer a different derivation, arriving at a new form of relative positional encodings, which not only has a one-to-one correspondence to its absolute counterpart but also enjoys much better generalization empirically (see Section 4). Firstly, in the standard Transformer (Vaswani et al., 2017), the attention score between query q_i and key vector k_j within the same segment can be decomposed as

$$\begin{aligned} \mathbf{A}_{i,j}^{\text{abs}} = & \underbrace{\mathbf{E}_{x_i}^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{E}_{x_j}}_{(a)} + \underbrace{\mathbf{E}_{x_i}^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{U}_j}_{(b)} \\ & + \underbrace{\mathbf{U}_i^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{E}_{x_j}}_{(c)} + \underbrace{\mathbf{U}_i^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{U}_j}_{(d)}. \end{aligned}$$

Following the idea of only relying on relative positional information, we propose to reparameterize the four terms as follows

$$\begin{aligned} \mathbf{A}_{i,j}^{\text{rel}} = & \underbrace{\mathbf{E}_{x_i}^\top \mathbf{W}_q^\top \mathbf{W}_{k,E} \mathbf{E}_{x_j}}_{(a)} + \underbrace{\mathbf{E}_{x_i}^\top \mathbf{W}_q^\top \mathbf{W}_{k,R} \mathbf{R}_{i-j}}_{(b)} \\ & + \underbrace{\mathbf{u}^\top \mathbf{W}_{k,E} \mathbf{E}_{x_j}}_{(c)} + \underbrace{\mathbf{v}^\top \mathbf{W}_{k,R} \mathbf{R}_{i-j}}_{(d)}. \end{aligned}$$

- The first change we make is to replace all appearances of the absolute positional embedding $\mathbf{U}_j$ for computing key vectors in term (b) and (d) with its relative counterpart $\mathbf{R}_{i-j}$. This essentially reflects the prior that only the relative distance matters for where to attend. Note that $\mathbf{R}$ is a sinusoid encoding matrix (Vaswani et al., 2017) without learnable parameters.
- Secondly, we introduce a trainable parameter $\mathbf{u} \in \mathbb{R}^d$ to replace the query $\mathbf{U}_i^\top \mathbf{W}_q^\top$ in term (c). In this case, since the query vector is the same for all query positions, it suggests that the attentive bias towards different words should remain the same regardless of the query position. With a similar reasoning, a trainable parameter $\mathbf{v} \in \mathbb{R}^d$ is added to substitute $\mathbf{U}_i^\top \mathbf{W}_q^\top$ in term (d).

- Finally, we deliberately separate the two weight matrices $\mathbf{W}_{k,E}$ and $\mathbf{W}_{k,R}$ for producing the content-based key vectors and location-based key vectors respectively.

Under the new parameterization, each term has an intuitive meaning: term (a) represents content-based addressing, term (b) captures a content-dependent positional bias, term (c) governs a global content bias, and (d) encodes a global positional bias.

In comparison, the formulation in Shaw et al. (2018) only has terms (a) and (b), dropping the two bias terms (c) and (d). Moreover, Shaw et al. (2018) merge the multiplication $\mathbf{W}_k \mathbf{R}$ into a single trainable matrix $\tilde{\mathbf{R}}$, which abandons the inductive bias built into the original sinusoid positional encoding (Vaswani et al., 2017). In contrast, our relative positional embedding $\mathbf{R}$ adapts the sinusoid formulation. As a benefit of the inductive bias, a model trained on a memory of some certain length can automatically generalize to a memory several times longer during evaluation.

Equipping the recurrence mechanism with our proposed relative positional embedding, we finally arrive at the Transformer-XL architecture. For completeness, we summarize the computational procedure for a N -layer Transformer-XL with a single attention head here. For $n = 1, \dots, N$:

$$\begin{aligned} \tilde{\mathbf{h}}_\tau^{n-1} &= [\text{SG}(\mathbf{m}_\tau^{n-1}) \circ \mathbf{h}_\tau^{n-1}] \\ \mathbf{q}_\tau^n, \mathbf{k}_\tau^n, \mathbf{v}_\tau^n &= \mathbf{h}_\tau^{n-1} \mathbf{W}_q^{n\top}, \tilde{\mathbf{h}}_\tau^{n-1} \mathbf{W}_{k,E}^{n\top}, \tilde{\mathbf{h}}_\tau^{n-1} \mathbf{W}_v^{n\top} \\ \mathbf{A}_{\tau,i,j}^n &= \mathbf{q}_{\tau,i}^{n\top} \mathbf{k}_{\tau,j}^n + \mathbf{q}_{\tau,i}^{n\top} \mathbf{W}_{k,R}^n \mathbf{R}_{i-j} \\ &\quad + \mathbf{u}^\top \mathbf{k}_{\tau,j} + \mathbf{v}^\top \mathbf{W}_{k,R}^n \mathbf{R}_{i-j} \\ \mathbf{a}_\tau^n &= \text{Masked-Softmax}(\mathbf{A}_\tau^n) \mathbf{v}_\tau^n \\ \mathbf{o}_\tau^n &= \text{LayerNorm}(\text{Linear}(\mathbf{a}_\tau^n) + \mathbf{h}_\tau^{n-1}) \\ \mathbf{h}_\tau^n &= \text{Positionwise-Feed-Forward}(\mathbf{o}_\tau^n) \end{aligned}$$

with $\mathbf{h}_\tau^0 := \mathbf{E}_{s_\tau}$ defined as the word embedding sequence. In addition, it is worth mentioning that a naive way to compute $\mathbf{A}$ requires computing $\mathbf{W}_{k,R}^n \mathbf{R}_{i-j}$ for all pairs (i, j) , whose cost is quadratic w.r.t. the sequence length. However, noticing that the value of $i - j$ only ranges from zero to the sequence length, we show a simple computation procedure in Appendix B, which reduces the cost to be linear w.r.t. the sequence length.

4 Experiments

4.1 Main Results

We apply Transformer-XL to a variety of datasets on both word-level and character-level language